RAJALAKSHMI ENGINEERING COLLEGE

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R2

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|---|---|
| **Started on** | Tuesday, 25 August 2026, 2:17 PM |
| **State** | Finished |
| **Completed on** | Tuesday, 25 August 2026, 2:44 PM |
| **Time taken** | 27 mins 1 sec |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include <stdio.h>

int main()
{
    int n;
    scanf("%d", &n);

    int c = 1;
    int outer = n - n / 2;

    int jcount = 0;
    int j = 1;

    while (j < n)
    {
        jcount++;
        j = j * 2;
    }

    c += outer + 1;
    c += outer * (jcount + 1);
    c += outer * jcount * (jcount + 1);
    c += outer * jcount * jcount;

    printf("%d", c);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4 | 30 | 30 | ✔ |
| ✔ | 10 | 212 | 212 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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